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ITECH3102 – Data Visualization
week 1

Lab Exercise – Introduction of Tableau Software

ID: _____Name: _____

Tableau software can produce interactive data visualization products based on different data. There are five main types of Tableau software products: Tableau Desktop, Tableau Server, Tableau Online, Tableau Reader, and Tableau Public. In this course, we employ two free products of Tableau software: Tableau Reader and Tableau Public.

Tableau Public

Tableau Public is a free application and visualization tool that allows general users to create and publish interactive data visualizations to the web. The Visualizations created by Tableau can be made available for a variety of applications. For example, visual results can be embedded into web pages, blogs and can be shared through social media or email.

Tableau Reader

Tableau Reader is a free application that lets general users to open and to view Tableau software files (called Tableau Workbook Files (.twbx)) that have been created by users

and any other users. This allows Tableau users to share visual results each other. Once opening and saving Tableau workbook files locally, users can view and interact with the visualization software. However, users of Tableau Reader cannot change the setting, parameter and design of existing Tableau workbook files.

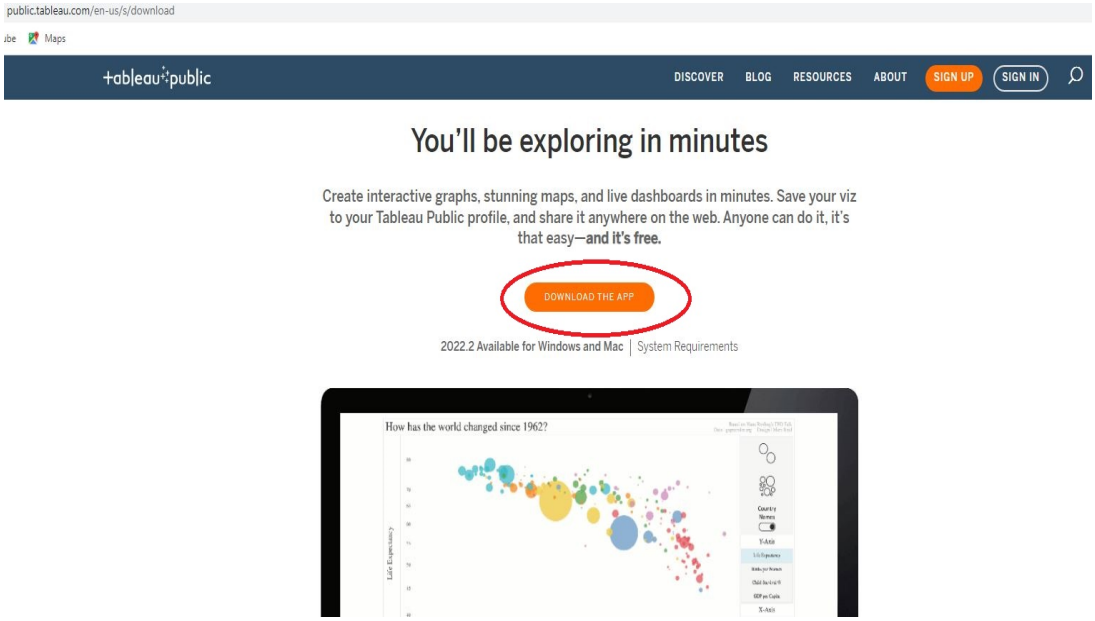
Project 1: Installation of Tableaus

1. Download Tableau Public.

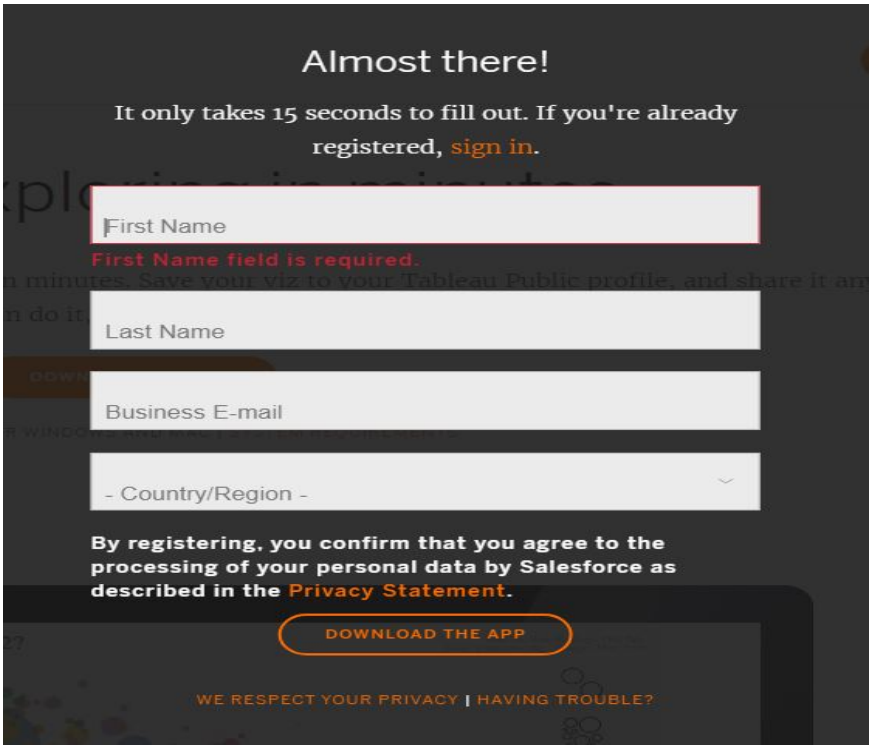
Basically, you can install free Tableau software on your computer: Tableau Public.

For downloading Tableau Public, go to the following website. You should install the Tableau software on your own machine.

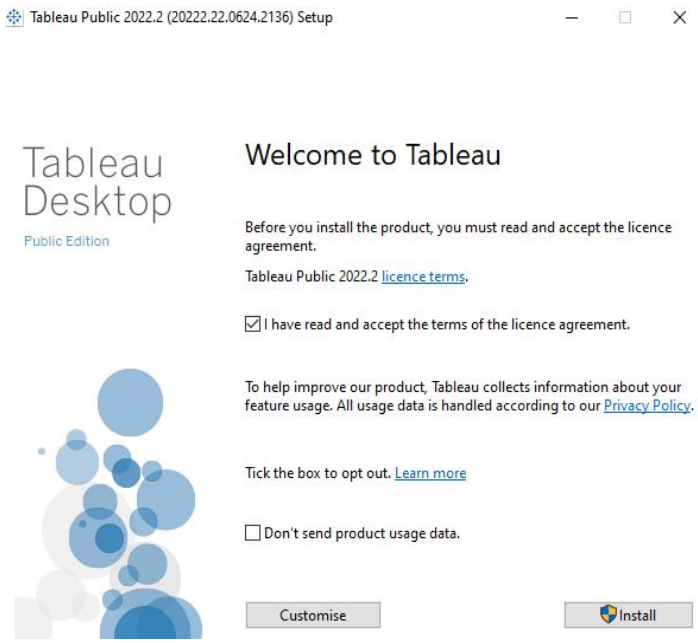
- **https://public.tableau.com/en-us/s/download**



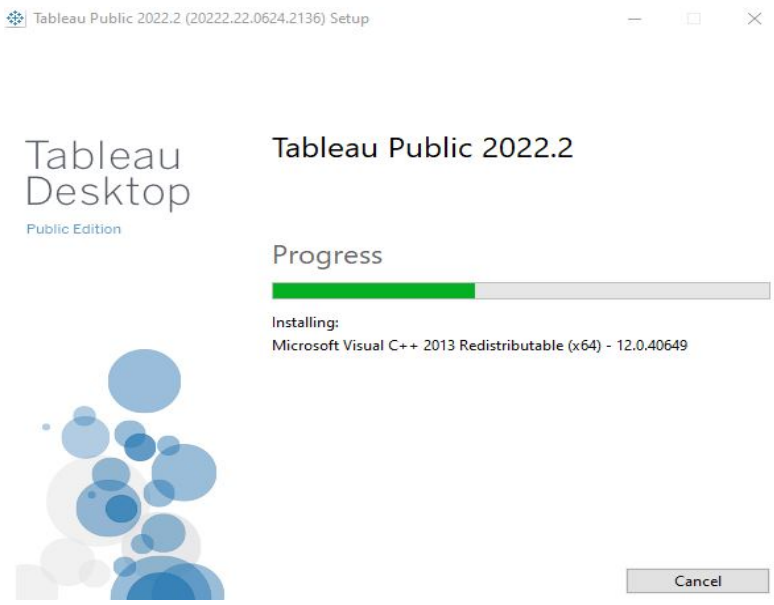
After clicking download button, you need to provide your information:



After providing your information and clicking download button, you start to download Tableau and then save it in your computer. Click downloaded TableauPublicDesktop-64bit-2022-2-0 to install it.



Click Install button to install software:



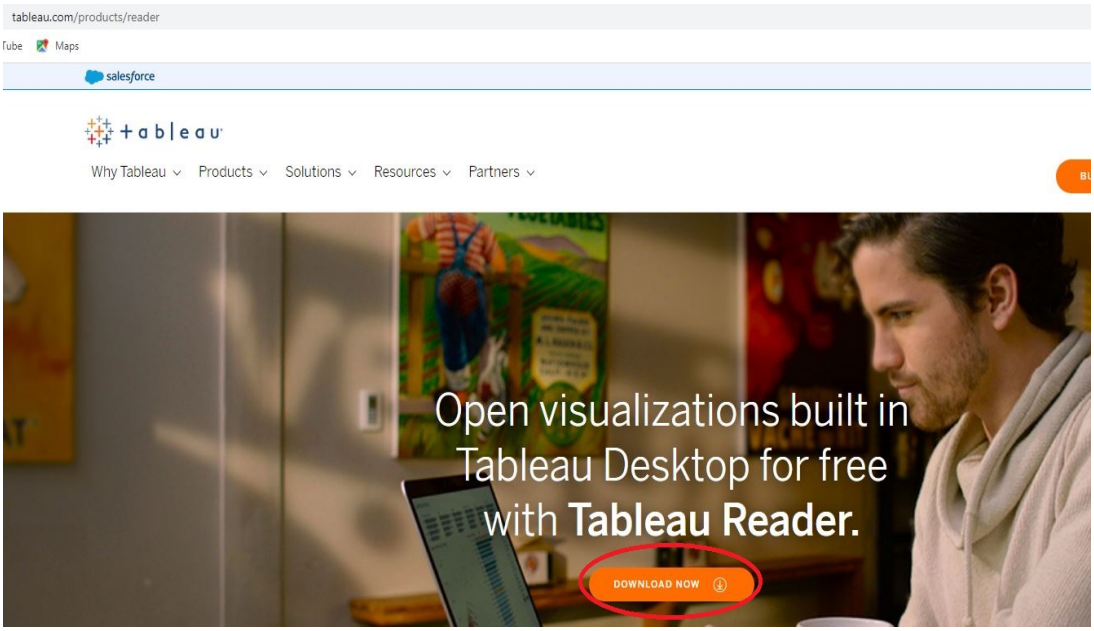
You'll find **Tableau icon** on your Desktop if you successfully install Tableau Public in your computer.



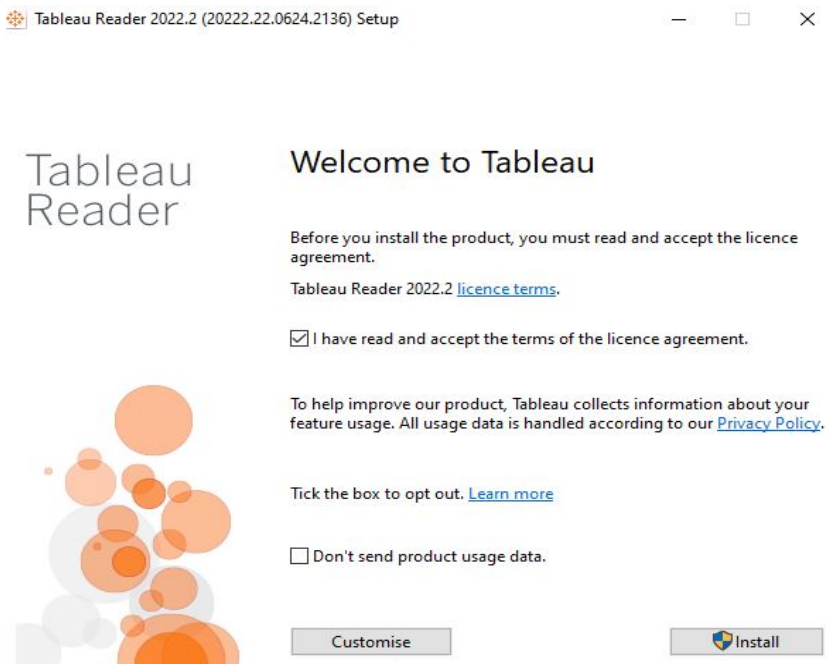
2. Download Tableau Reader

For downloading Tableau Reader, go to the following website. You should install the Tableau Reader on your computer.

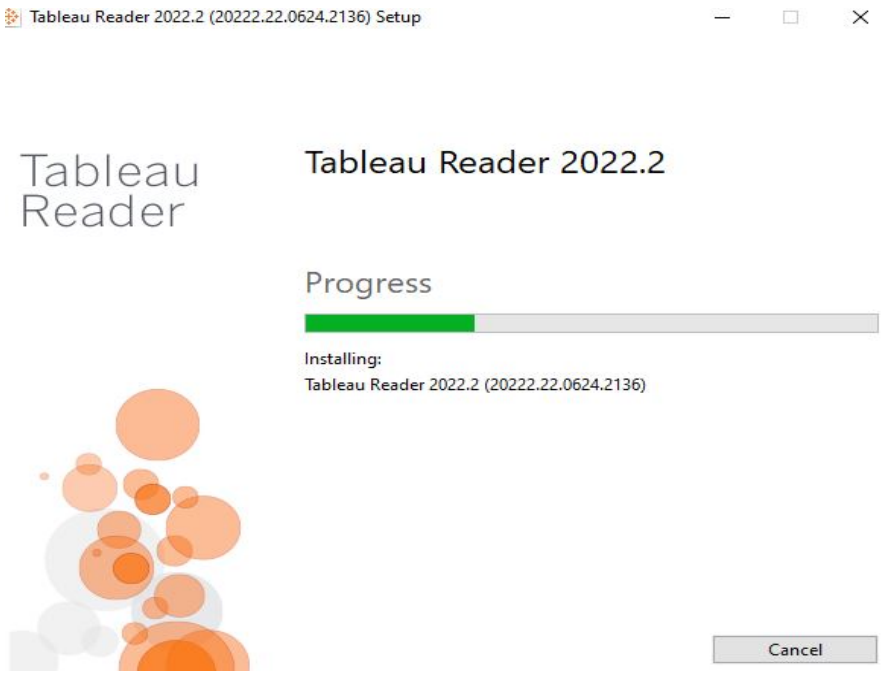
Tableau Reader:
https://www.tableau.com/products/reader



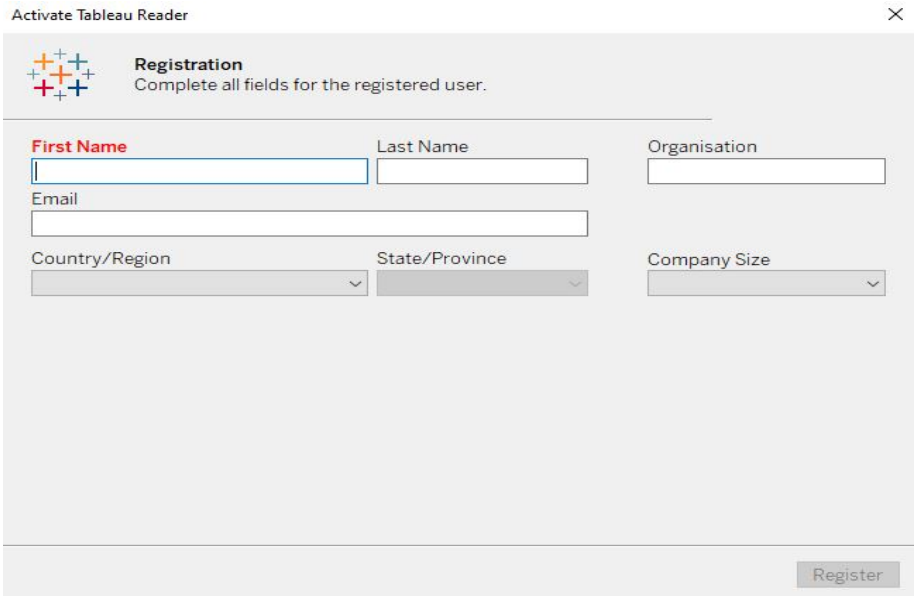
After clicking download button, you start to download Tableau Reader and save it in your computer. Click downloaded TableauReader-64bit-2022-2-0 to install it.

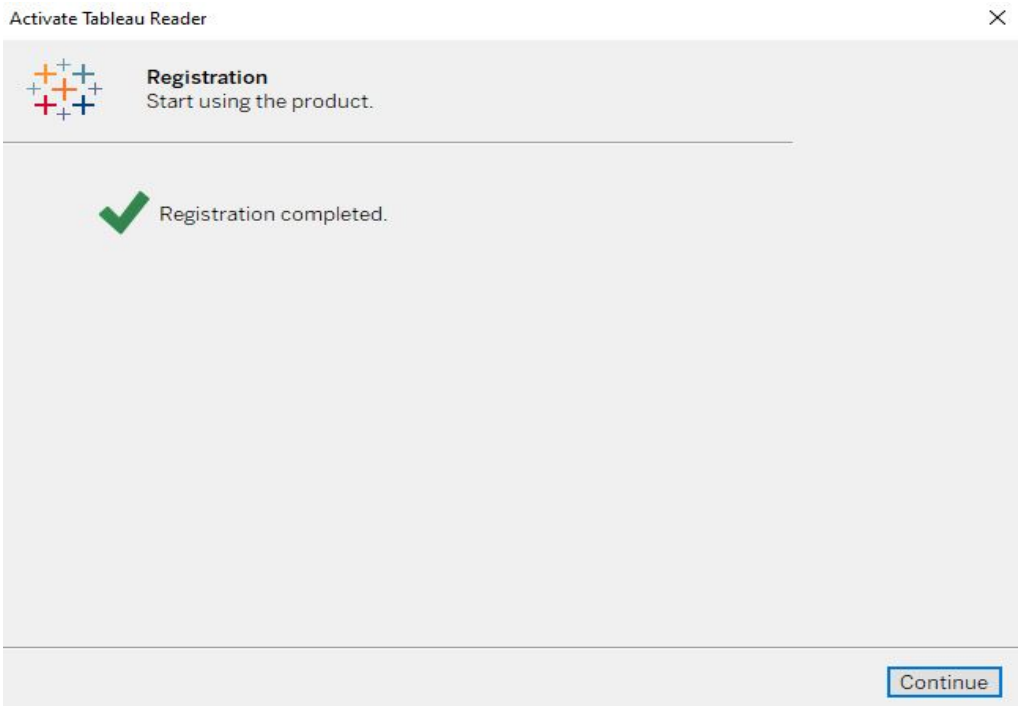


Click Install button to install software:



You need to provide your information to activate Tableau Reader



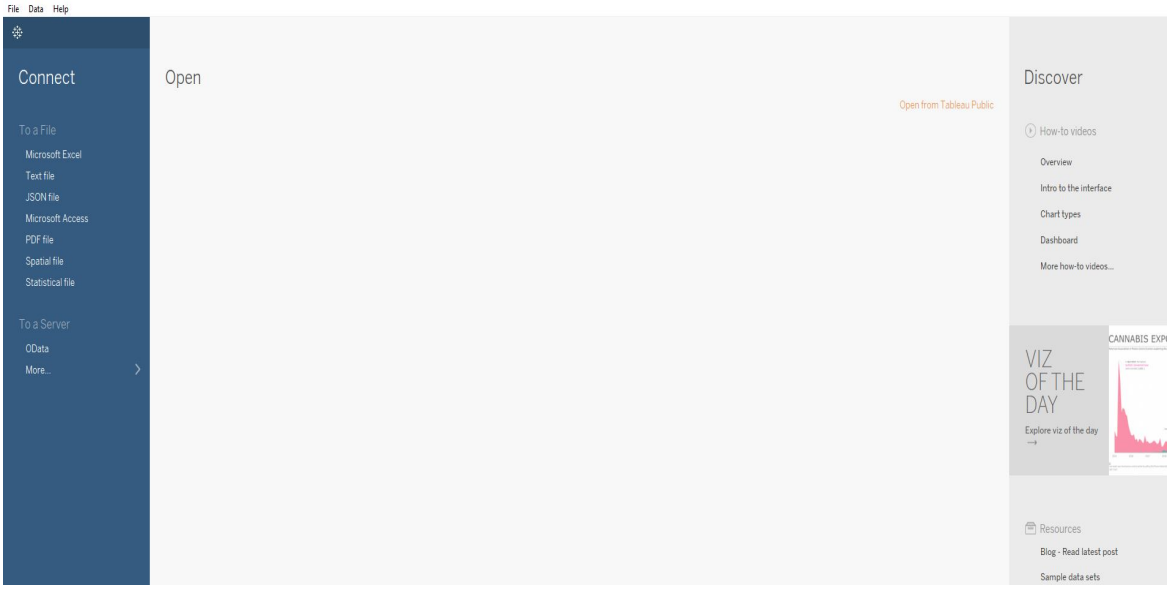


Project 2: Understand user interface of Tableau

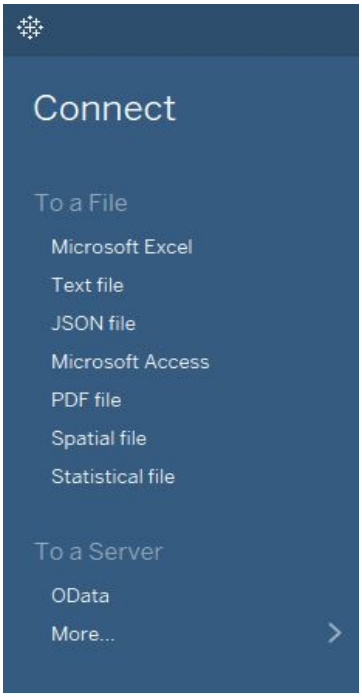
When clicking Tableau logo, you will go to a Start page.

Start page

The start page consists of three panes: left, center and right panes.

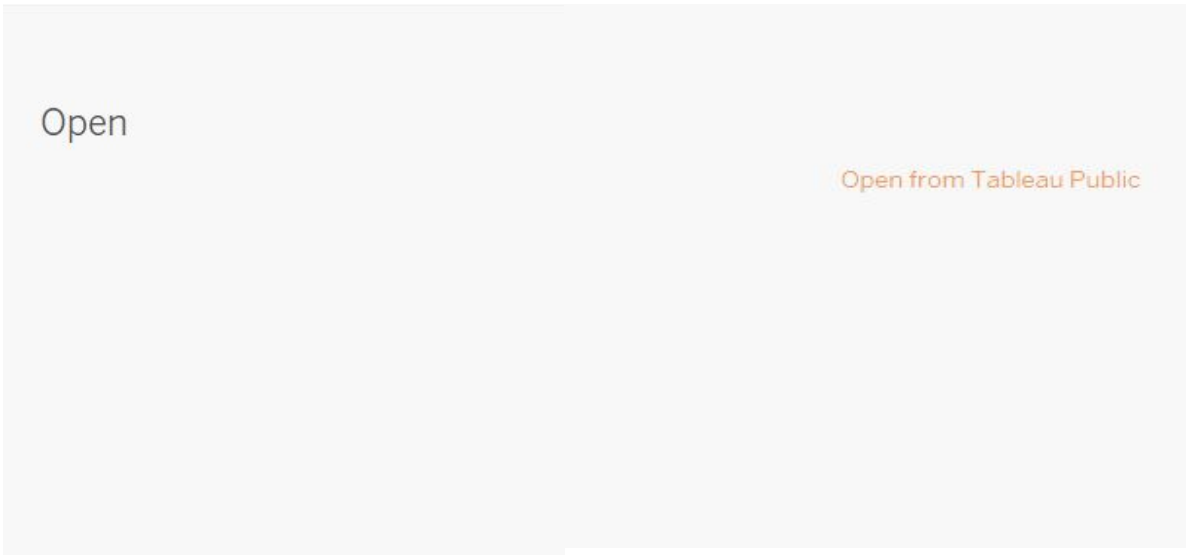


Left pane: **connect pane**.



In the connect pane, users can connect to several kinds of data sources. The connect pane provides convenient access to data stored in files such as Excel, Text and JSON files, etc., on your computer.

Centre pane: **open pane**.



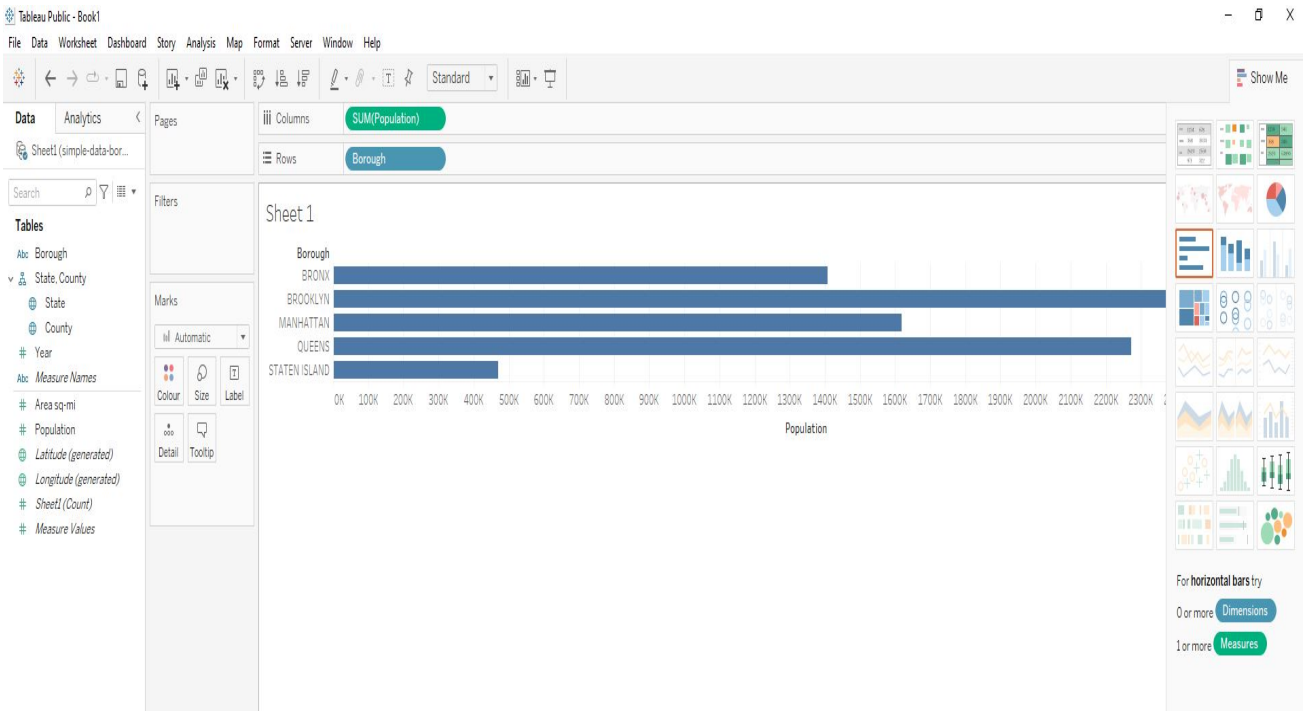
The open pane is in the center of the screen. In the open pane, users can open the work that users recently saved or opened. If you have just installed Tableau, the white space to the right of the connect pane will be empty on your screen. Users can also use open pane to connect some samples provided by Tableau.

Right pane: discover pane.

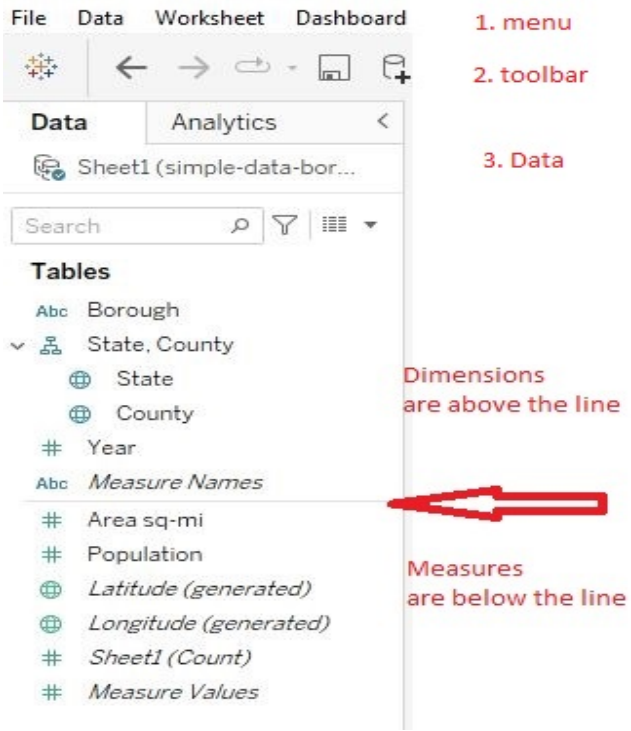


On the right side of the start page, you will find online learning resources that are provided by Tableau. In the discover pane, users can find some training examples, videos and users forum, etc.

Tableau's primary interface



On the left of Tableau's primary interface



- The **menu** provides various items for performing a wide range of functions.
- The **toolbar** allows common functions such as undo, redo, save, add a data source, and so on.
- The **Data** pane is active when the Data tab is selected and lists all tables and fields of the selected data source.
- The **Analytics** pane is active when the Analytics tab is selected and gives options for supplementing visualizations with visual analytics.

Dimension and Measure

The fields from the data source are visible in the Data pane and are divided into Measures and Dimensions. Tableau will classify each field as a dimension or a measure when you connect to data. Tableau will then group the fields by their dimension or measure classification on the lefthand side of the workspace.

In the Data pane, each table will have Measures and Dimensions separated by a line:

- **Dimensions.** A dimension is a field that can be considered an independent variable based on Tableau's Knowledge Base. By default, Tableau treats any field containing qualitative, categorical information as a dimension. For example, **categorical data** can be strings, date, categorical on which we can group or categorize
    - Measure Names
      abc before this field indicates that it represents the names of the fields

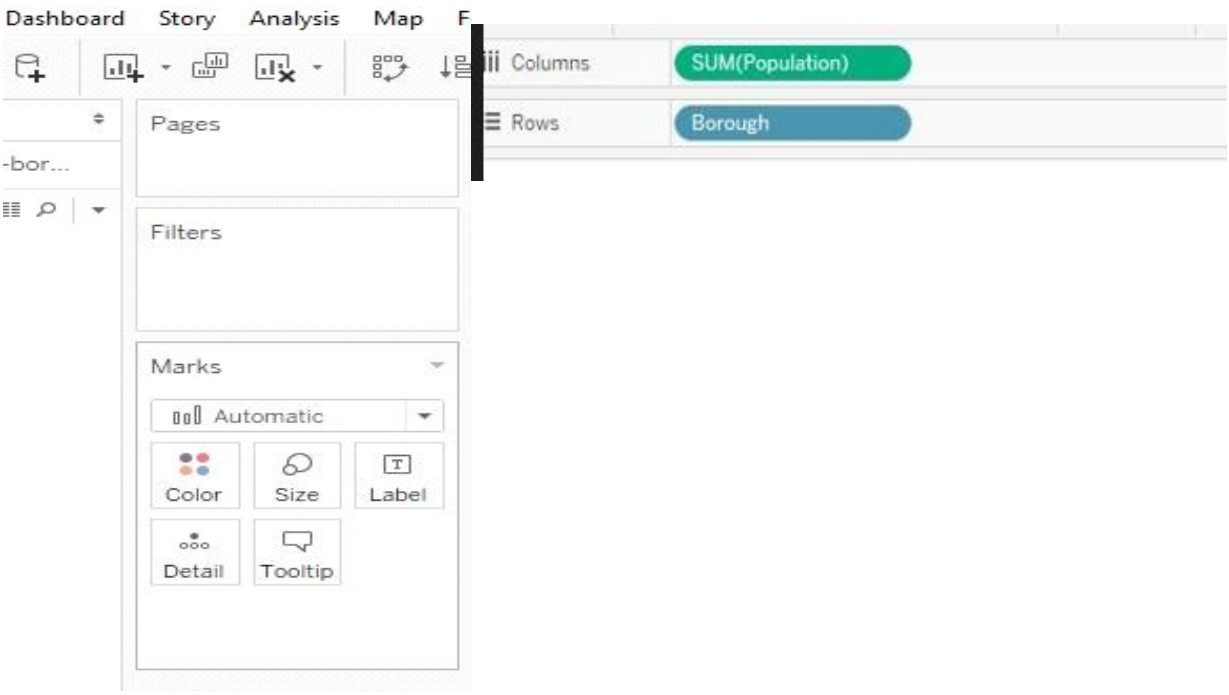
- **Measures.** A measure is a field that is a dependent variable; that is, its value is a function of one or more dimensions. Tableau treats any field containing numeric (quantitative) information as a measure**.** In other words, measures are values that are aggregated. For example, for **quantitative data such as population,** we can do math operation such as summing or averaging or getting minimum/maximum, etc.
    - Number of Records
      # before this field indicates that it is a numeric filed, which is used to count number of entries in data source



For example, in above picture, the Population field is used as a measure in this view. Specifically, it is being aggregated as a sum.

The Borough field is a dimension with one of five values for each record of data: BRONX, BROOKLYN,…, or STATEN ISLAND.

Various shelves



Various shelves such as Pages, Columns, Rows, and Filters serve as areas to drag and drop fields from the data pane. Three main components on the various shelves are Pages shelf, Filters shelf and Marks card.

- **Pages shelf** works like a filter. The Pages shelf allows you to "flip" through a "page" for each dimension member and/or add animation to a view. It will hide elements on the sheet that are unrelated to currently selected page.
- **Filters shelf.** Any dimension or measure that you filter a view by will be displayed here. Filters shelf will filter your categorical data (dimensions), quantitative data (measures), or date fields. When you drag a field from Dimensions or Measure to the Filters shelf, Filters shelf opens a box where you can define the filter.
- **Marks card.** Each square in this area is called a Marks Card, which are called that because they influence the marks on the view. Marks card will allow users to edit color, change size, add label, show detail and show tooltip for marks used in the sheet.
- **Columns Shelf:** Fields placed here will create columns on the view.
- **Rows Shelf**: Fields placed here will create rows on the view.

On the center of Tableau's primary interface

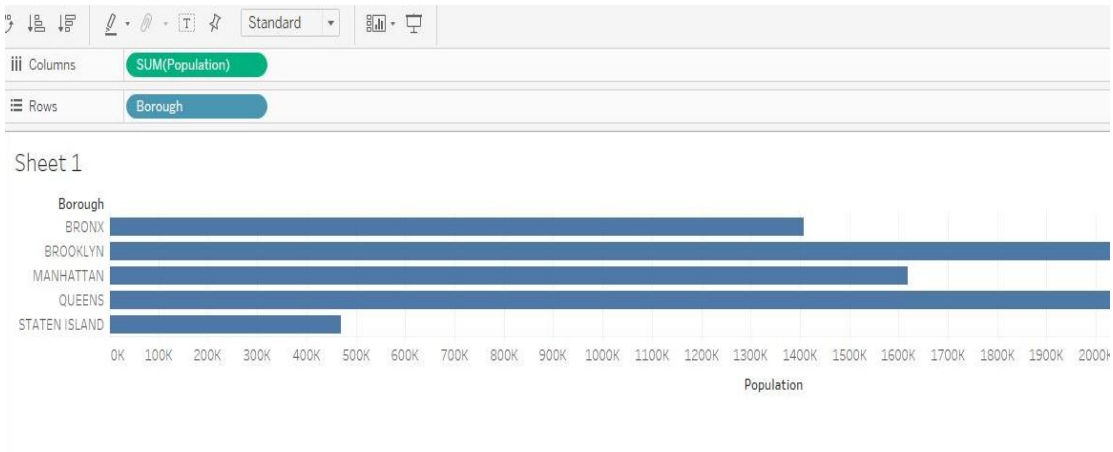
Two main workspaces

After connecting data source to your computer, Tableau software provides two main workspaces: **sheet** and **dashboard** where you can build your visualization.

Tableau provides users with two main workspaces: **sheets** and **dashboards** to develop visual projects. These two workspaces provide **visual building environments** where users are able to build and develop visual projects by using different kinds of controls, tools and menus that Tableau provides.

(1) Sheets

A sheet is used to build **individual project** of data visualization. The interface of a sheet is shown as following.

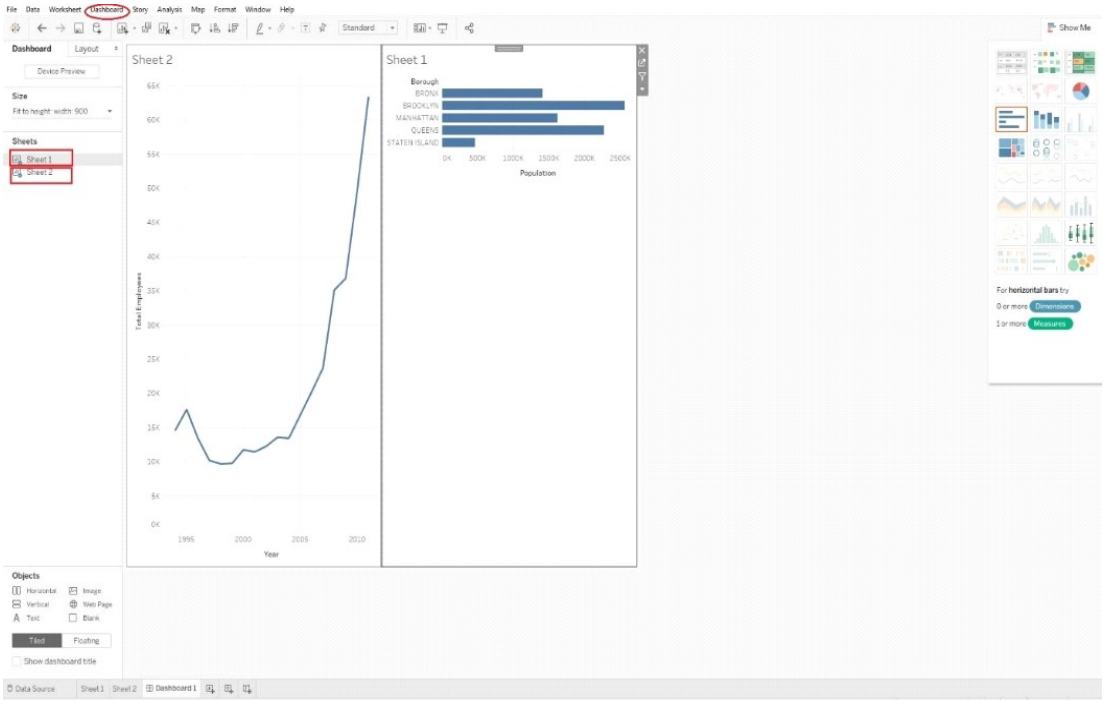


The canvas or view is where Tableau will draw the data visualization. The area that displays a data visualization is a view. In addition to dropping fields on shelves, you may also drop fields directly on to the view. A title is located at the top of the canvas. By default, the title displays the name of the sheet, but it can be edited or hidden.

(2) Dashboard

Dashboard can include several individual sheets, where each sheet shows different views of data. Users can create dashboards by primarily using a drag-and-drop operation so that only little knowledge of programming is required.

In addition, when individual sheet is placed into the dashboard workspace, interactions with data elements on one sheet can be made with the data element on other sheets. Therefore, designer can design dashboards by using different kinds of interactivity techniques to effectively communicate to audience. In this course, you will design your dashboards by combining other interactivity techniques to implement your final project.



On the right of Tableau's primary interface



Tableau provides two main approaches to visualize data: **Drag-drop** and **Show Me** card

- **Drag and drop**. You just drag related fields from Dimensions or Measures to Rows shelf or Columns shelf and drop them. Tableau will automatically visualize your data.
- **Show Me card**. When you click Show Me in the upper-right corner of the authoring interface, you will see thumbnails for 24 different chart types. In this way, when you select multiple fields in Dimensions or Measures by using Control key and then click Show Me card, Show Me card will recommend suitable visual charts that are highlighted for your choices. After you select a chart type, Tableau will visualize your data.
- **Show Me card** provides a nice shortcut to creating several useful chart types, **but this course will primarily focus on creating charts manually, which allows you to gain a better understanding of how each visualization in Tableau is created.**

Project 3 Getting data into Tableau

In this project, we learn how to get data into Tableau. We can get data from local computer, database, servers, and from public data sources in the cloud. In this course, we focus on connecting data files from local computer.

Download data set: **simple-data-bogoughs.xlsx** from week 1 in Moodle

Task: The given dataset about information in a city is stored in local computer. The goal of this task is to get data into Tableau and understand basic data types.

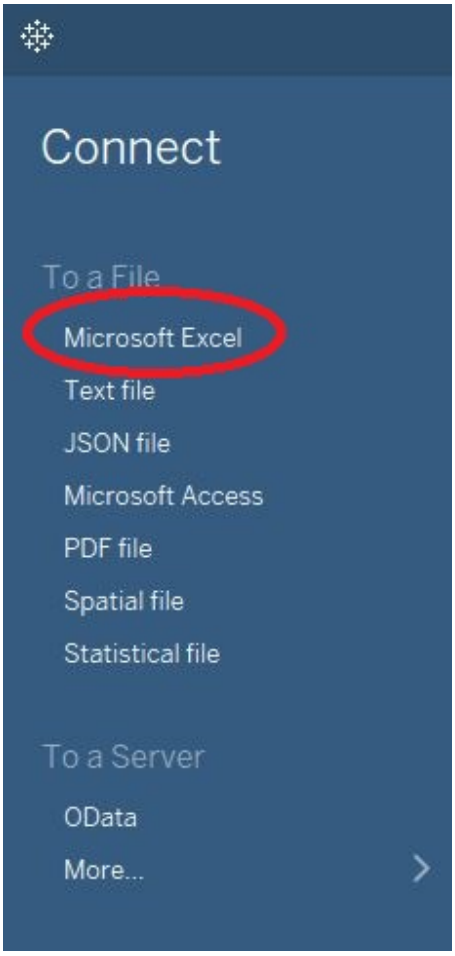
Step 1. Start up Tableau
Step 2. Connect to data source
Step 3. Preview data
Step 4. Understand basic data types and operations

Step 1. Start up Tableau

Double clicking **Tableau icon** would open the window of Tableau. On the left panel of windows, several links are provided to allow users to connect different kinds of data

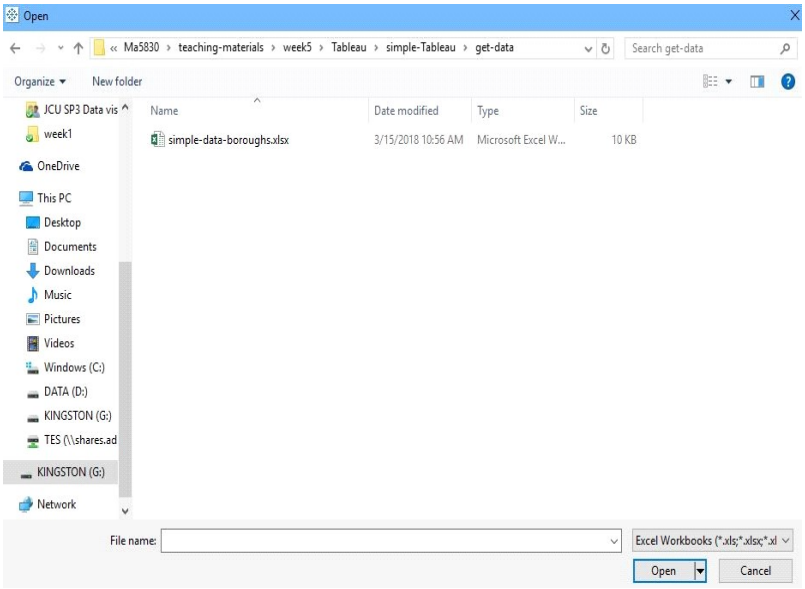
sources. These data sources include Microsoft Excel, text file, JSON file, Microsoft Access, etc.

In this course, we use Tableau Public that only allows users to connect to Excel, Access, comma-delimited files, etc. In addition, Tableau Reader can only open and view packaged software file Tableau workbooks (**.twbx**).



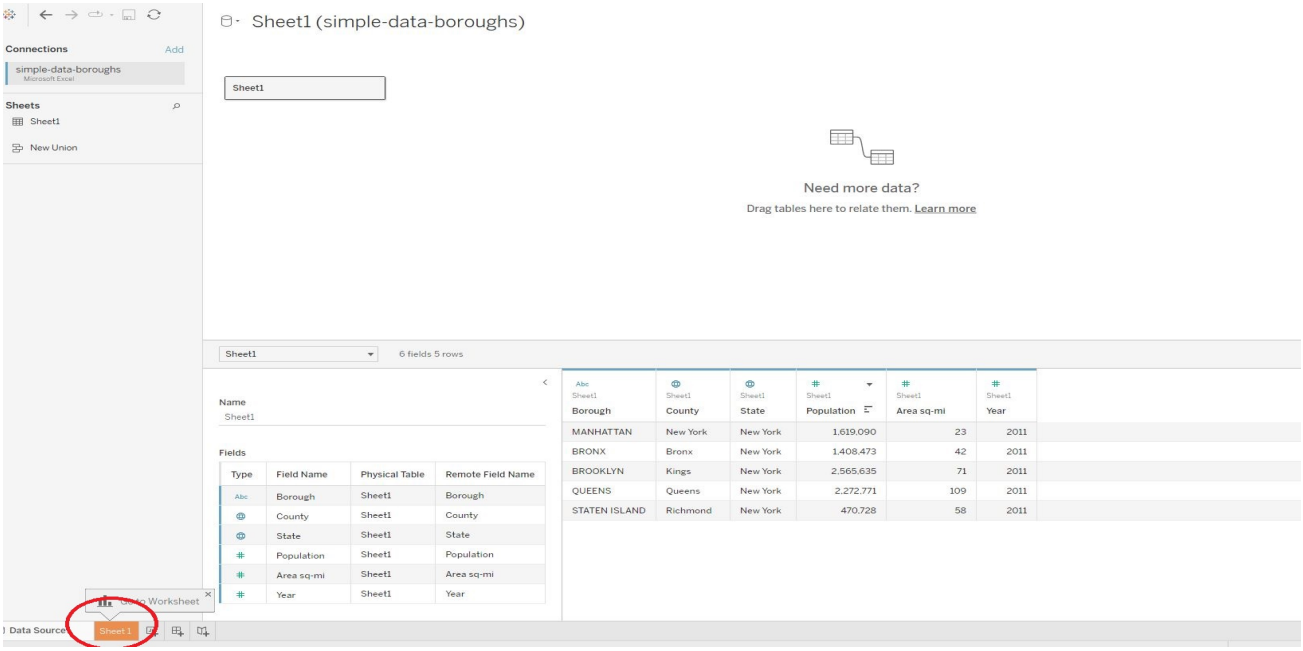
Step 2. Connect to data source

Navigate to the Microsoft Excel and **click** it to open a window. **Select** the required data file **simple-data-bogoughs.xlsx** and connect it to Tableau.



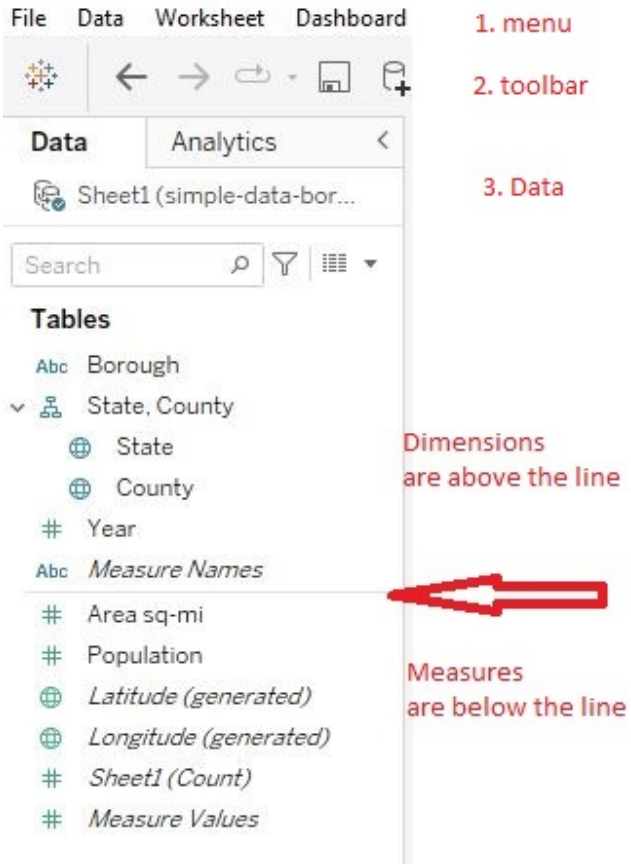
Step 3. Preview data

Clicking **Open** button will show a tabular view of data in the Preview area in the bottom half of the page(window). This allows users to see data before users place the sheet on left data pane.



Step 4. Understand basic data types and symbols operations

Navigate to the **Worksheet** tab at the bottom of window. After clicking **Sheet 1** tap on the bottom, your data will be connected to your computer and shown in data pane on the left of interface. Tableau shows two classified data fields: Dimensions and Measures.



Color encoding

In Tableau, **blue color** is used to stand for discrete data such as **strings, date, categorical** on which we can group or categorize.

Green color is used to stand for continuous data on which we can do math operation such as summing or averaging.

Symbol usage:

Symbol # to the left field names indicates **continuous data** types.
Globe symbol next to the field name indicates **geographic data** types.

Project 4. Getting data out of Tableau (save as a data source)

In this project, we learn how to get data out of Tableau.

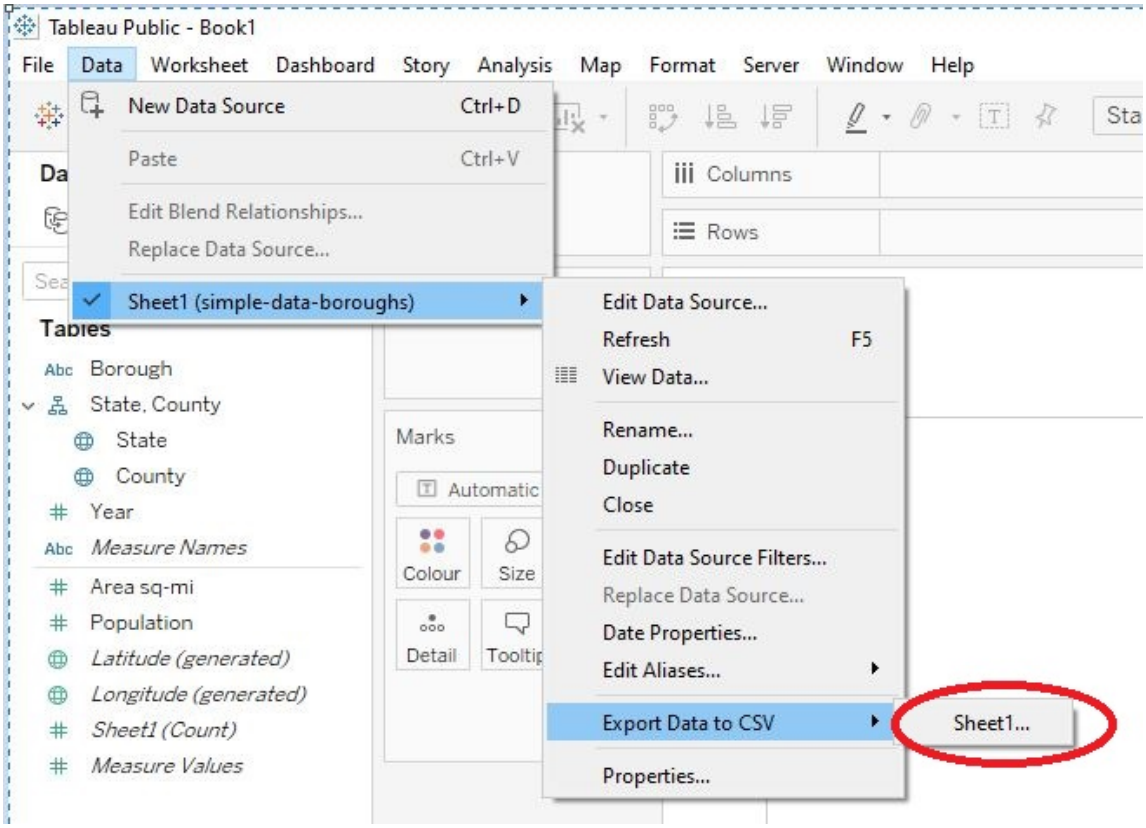
Download data set: **simple-data-bogoughs.xlsx** from week 1's lab in Moodle

Task: given dataset about information in a city is stored in local computer. The goal of this task is to get data out of Tableau and understand basic data types.

Step 1. Navigate to data source in data pane
Step 2. Select data export option
Step 3. Export data

Step 1. Navigate to data source in data pane

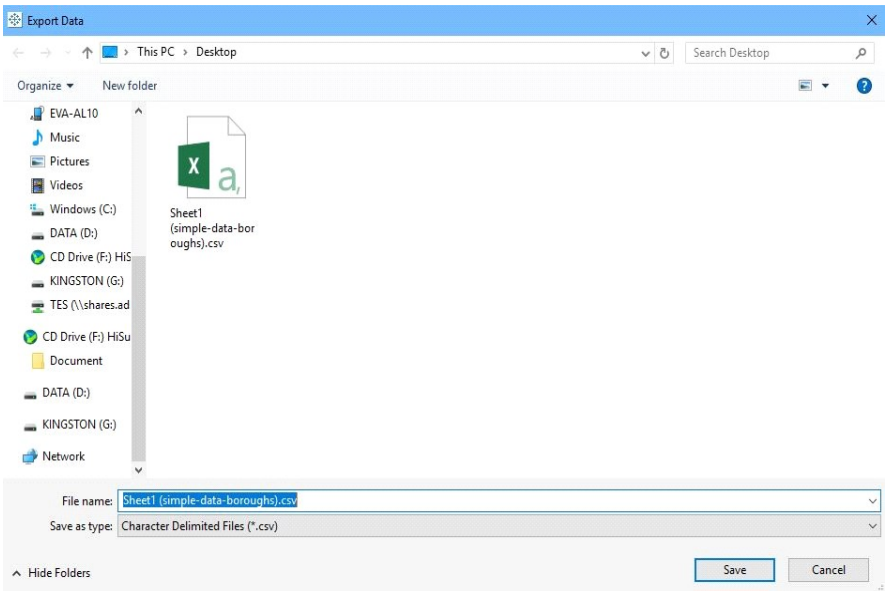
- Clicking **Sheet 1** tab at the bottom of page takes you to worksheet.
- **Navigate** to Data at the upper part of the page and point to Sheet 1(**simple-data-boroughs**), then point Export Data to CSV and point Sheet 1


Tableau Public - Book1
File
Data
Worksheet
Dashboard
Story
Analysis
Map
Format
Server
Window
Help
New Data Source Ctrl+D
Paste Ctrl+V
Edit Blend Relationships...
Replace Data Source...
Sheet1 (simple-data-boroughs)
Edit Data Source...
Refresh F5
View Data...
Rename...
Duplicate
Close
Edit Data Source Filters...
Replace Data Source...
Date Properties...
Edit Aliases...
Export Data to CSV
Sheet1...
Properties...
Columns
Rows
Tables
Borough
State, County
State
County
Year
Measure Names
Area sq-mi
Population
Latitude (generated)
Longitude (generated)
Sheet1 (Count)
Measure Values
Marks
Automatic
Colour
Size
Detail

Step 3. Export data

Click the **Sheet 1** to open an Export Data window**.**

Select a folder where you save your data file with .csv extension. Now you have changed your data from .xlsx format to .csv format.



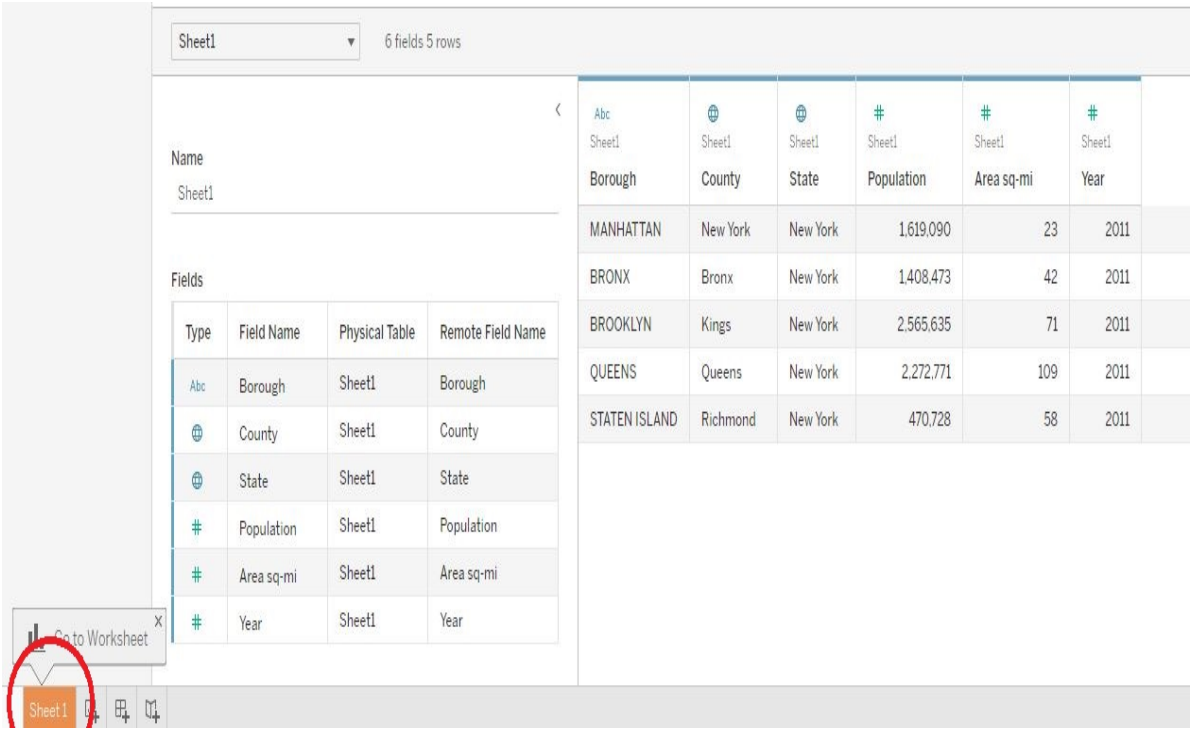
Project 5. Getting data out of Tableau 2 (save as a workbook)

Task: The given the work you have already done in your workbook. The goal of this task is to save your work as .twbx file that you can run later by using Tableau Reader.

Download Dataset: simple-data-bogoughs.xlsx from week 1's lab in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly
Step 2 Create a bar chart
Step 3. Save the project from File
Step 4 Sign in and save the project
Step 5 Download your project

Step 1. **Load the data into Tableau** and check that your data has been loaded correctly

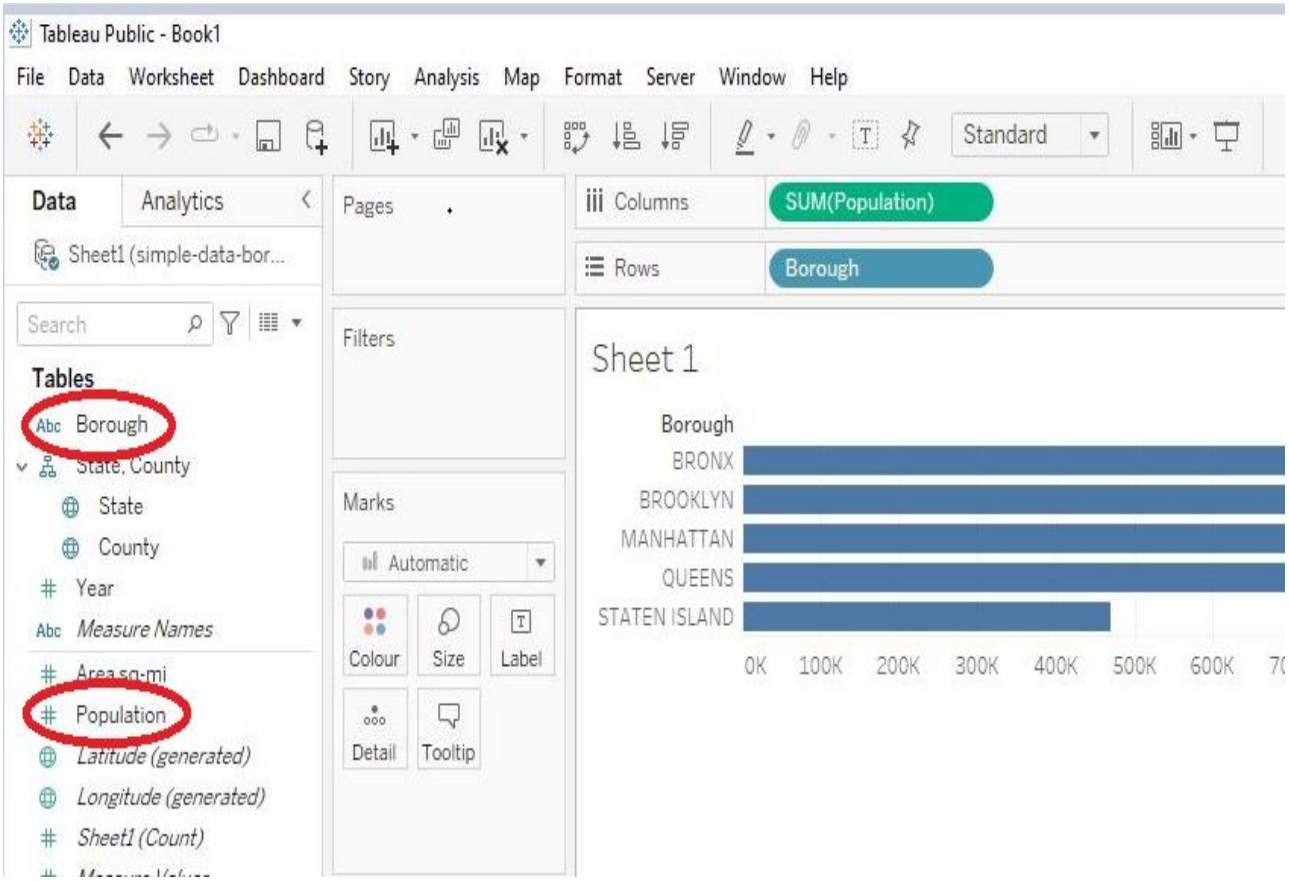


After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create a bar chart

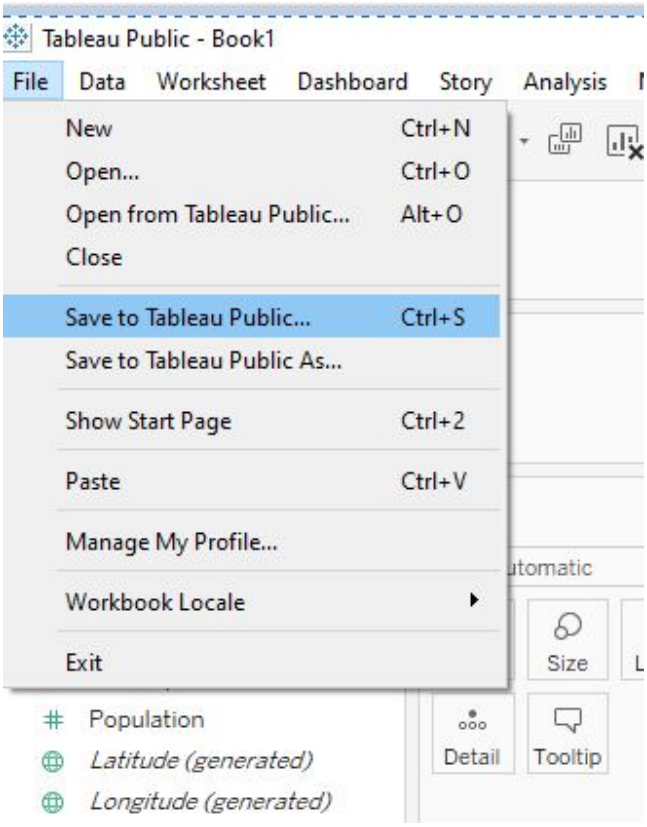
Drag and drop **Population** field from the Measures section in the data pane on to Columns shelf. You now have a bar chart with a single bar representing the sum of population for all of the data in the data source.

Drag and drop **Borough** field from the **Dimensions** section in the data pane on to the Rows shelf. This slices the data to give you five bars, each having a length that corresponds to the sum of population for each area.



Step 3. Save the project from File

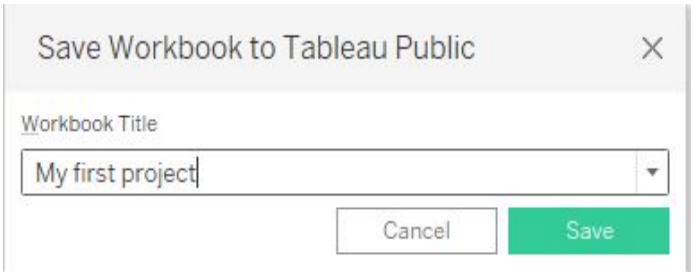
Click **File** from the menu bar at top and select **Save** to Tableau Public



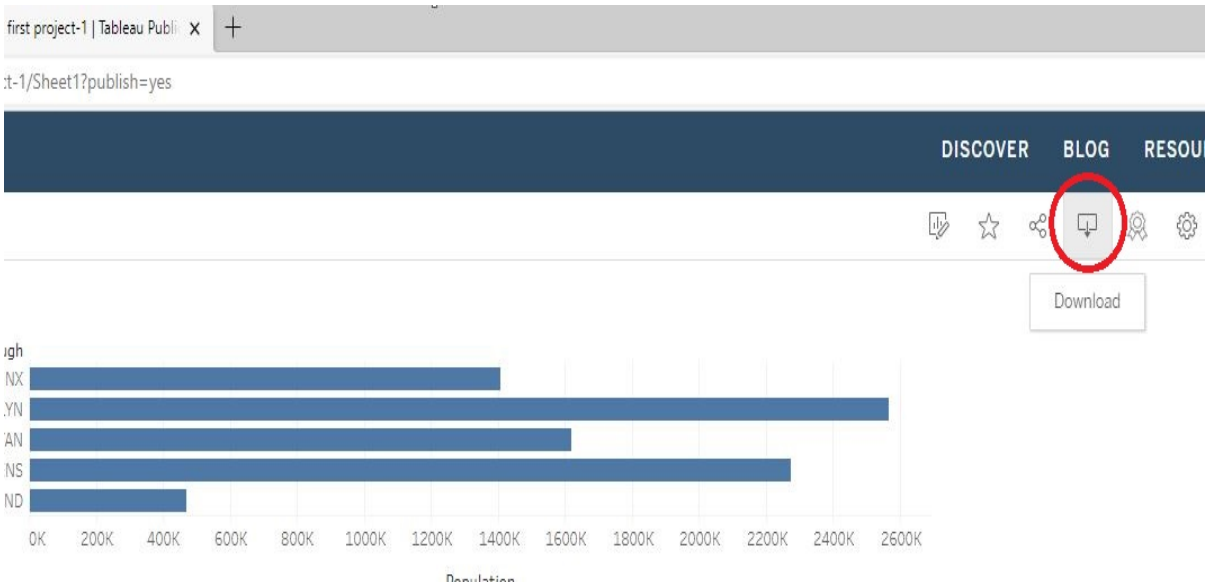
Step 4. Sign in and save the project



You need to provide your email to save the project. After providing your email and password, click Sign In, you'll get:

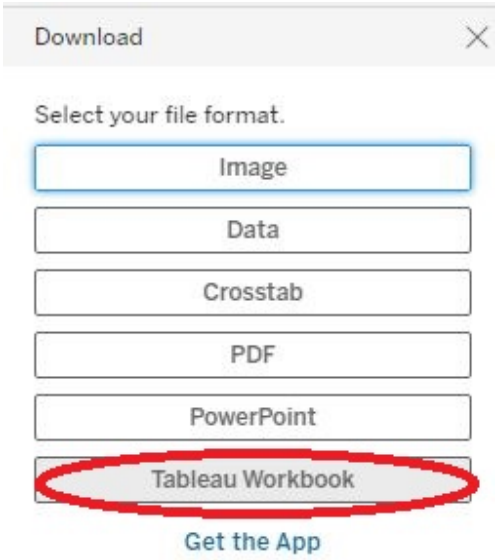


Give a title to your project, then click Save button and you'll get

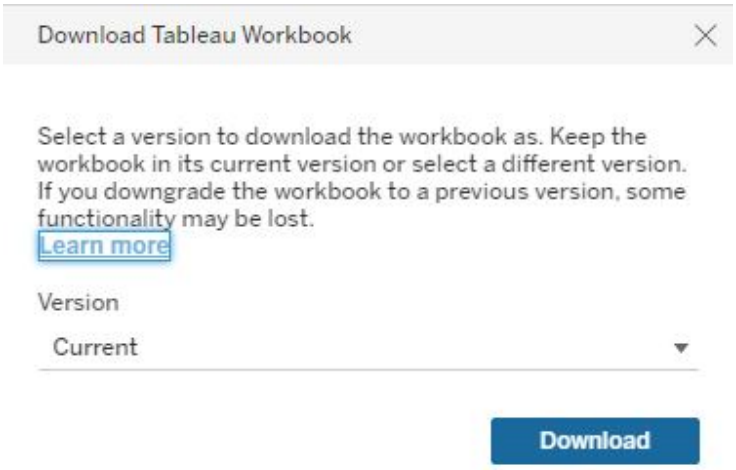


Step 5. Download your project

From Tableau Public website, click Download tap to download your project.



Then select Tableau Workbook as file format



And then click **Download**, and your work will be saved as My first project.twbx in your computer. You can run My first project.twbx later by using Tableau Reader.



Project 6. Answering questions (Please do this at your home by using your own computer)

1. **Install** Tableau Public and Tableau Reader in your computer.

2. Watch the following video and summarize your comment.

   https://www.ted.com/talks/hans_rosling_the_truth_about_hiv?referrer=playlist-the_best_hans_rosling_talks_yo

3. Explore the Discover Pane and find online learning resources such as examples and videos that are provided by Tableau.

Project 7. Create a Professional Report

Summarize the above experiments procedures, results, answers for questions and screenshots (project 1, 2, 3, 4,5, 6) into one report. Your report is the assignment task that is required to be submitted for evaluation on week 4. Create a report by the following steps.

Please note that

- as you may have a different version of Microsoft Word, **the following steps are just for your references.**

1. Open Word document, record and copy your OWN experiments procedure, results, answering questions and screenshots (from project 1, 2, 3, 4) into a word document. Please note that you cannot copy the screenshots provided in this lab sheet and paste them to your report. Otherwise, your report will not be marked.
2. Select **all content** , align all text to both **left and right** margin.
3. On the **Home** tab, in the Styles group, right-click **Heading 1** in the Quick Style gallery. Click **Modify** style , click **Format** , click **Font** , then select **Font color** to Black, select **Font style** to Bold, select **Font size** to 18 , then click Ok
4. **Create a shortcut key for Chapter**. Click **Format** , click **Shortcut key** , in text box **Press new shortcut key**, type **Ctrl + A**, click **Close**, click **OK**
5. **Select a style for sub-Chapter**. On the **Home** tab, in the **Styles** group, right-click **Heading 2** in the Quick Style gallery. Click **Modify** style , click **Format** , click **Font** , then select **Font color** to Black, select Font style to Italic, select Font size to 14 , then click Ok

6. **Create a shortcut key for sub-Chapter**. click **Format** , click **Shortcut key** , in text box **Press new shortcut key**, type **Ctrl + B,** click **Close**, click **OK**
7. Insert a page number to your document
8. You may call the summary of Week 1 lab work as Chapter 1 and the summary of Week 2 lab work as Chapter 2, ..., etc.
9. **Format your Chapter 1**. At the beginning in your report, Press shortcut key **Ctrl + A ,** then type **Chapter 1: Introduction of Tableau Software**
10. **Format your sub-Chapter**. At proper position within Chapter 1, Press shortcut key **Ctrl + B** , then type 1.1 **Project 1: Installation of Tableaus**
11. Repeat step 10 and format sub-Chapter for Project 2 and Project 3
12. **Insert a table of contents for your report**. At the beginning of your report, **Click** where you want to insert the table of contents
13. On the **References** tab, in the **Table of Contents** group, click insert **Table of Contents**, then select Formal , click OK
14. **Insert a cover page**. On the **Insert** tab, in the **cover page** group, select **Conservative,** then add your name and ID number.